

DUAL AXIS LIGHT TRACKING SOLAR PANEL

Principle: -

Our project is a Dual-Axis Solar Tracking System that automatically changes the position of the solar panel to face the Sun directly. This helps the panel receive maximum sunlight and produce higher power.

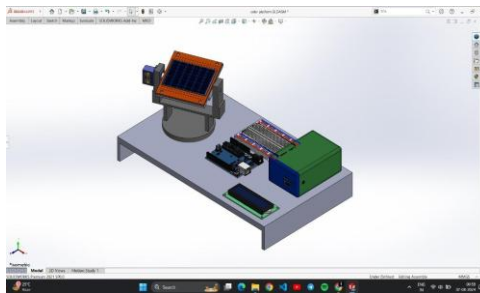
Mechanism: -

- We use four LDRs (Light Dependent Resistors) as light sensors.
- The LDRs are placed at the four corners of the solar panel.
- They detect the intensity of sunlight from different directions.
- Based on the difference in light intensity, servo motors rotate the panel horizontally and vertically.
- The panel keeps moving until the light is balanced on all four LDRs, meaning the panel is directly facing the Sun.

Functions: -

- Maximizes solar panel efficiency by keeping it aligned with the Sun.
- Provides dual-axis movement, allowing the panel to track the Sun throughout the day and across different seasons.
- An LCD display shows the solar panel's real-time voltage output.

CAD: -



CIRCUIT: -

